

SIMATS ENGINEERING
ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

Code of Conduct:

I Prethivika.G(192521085) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Prethivika G

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

1.E-Commerce Admin Dashboard (BTL: BL5 – Evaluate)

Design a Swing-based GUI application for an e-commerce platform where administrators can manage products, orders, and customers. Use AWT/Swing components such as tables, buttons, text fields, and menus to perform operations like adding products, updating inventory, and viewing orders. Implement event listeners for all user actions. Ensure proper validation for product details and stock updates. Evaluate the effectiveness of different layout managers in handling complex dashboards. Analyze system performance when handling large datasets and suggest improvements for scalability and usability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

PROGRAM:

Q1 E-Commerce Admin Dashboard (BTL: BL5 – Evaluate)

20 marks

Design a Swing-based GUI application for an e-commerce platform where administrators can manage products, orders, and customers. Use AWT/Swing components such as tables, buttons, text fields, and menus to perform operations like adding products, updating inventory, and viewing orders. Implement event listeners for all user actions. Ensure proper validation for product details and stock updates. Evaluate the effectiveness of different layout managers in handling complex dashboards. Analyze system performance when handling large datasets and suggest improvements for scalability and usability.

```
import javax.swing.*.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*.*;
import java.awt.event.*;
/*
<applet code="EcommerceDashboard.class" width="1150" height="750">
</applet>
*/
public class EcommerceDashboard extends JApplet {
    private DefaultTableModel productModel;
    private DefaultTableModel orderModel;
    private DefaultTableModel customerModel;
    private JTable productTable;
    private JTable orderTable;
    private JTable customerTable;
    private JLabel revenueLabel;
    private JLabel activeLabel;
    private JLabel stockLabel;
    private Color dark = new Color(30, 30, 36);
    private Color panel = new Color(43, 43, 50);
    private Color white = new Color(240, 240, 245);
    private Color green = new Color(46, 204, 113);
    private Color blue = new Color(52, 152, 219);
    private Color red = new Color(231, 76, 60);
    public void init() {
        setLayout(new BorderLayout());
        getContentPane().setBackground(dark);
        JMenuBar menuBar = new JMenuBar();
        JMenu menu = new JMenu("Dashboard System");
        JMenuItem export = new JMenuItem("Export Dashboard Data");
        menu.add(export);
        menuBar.add(menu);
        setJMenuBar(menuBar);
        export.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JOptionPane.showMessageDialog(EcommerceDashboard.this, "Dashboard data
compiled successfully.", "Export Complete", JOptionPane.INFORMATION_MESSAGE);
            }
        });
    }
}
```

```

});
JPanel kpiPanel = new JPanel(new GridLayout(1, 3, 20, 10));
kpiPanel.setBackground(dark);
revenueLabel = new JLabel("$0.00", SwingConstants.CENTER);
activeLabel = new JLabel("0 Active", SwingConstants.CENTER);
stockLabel = new JLabel("0 Items Low", SwingConstants.CENTER);
revenueLabel.setFont(new Font("Arial", Font.BOLD, 24));
activeLabel.setFont(new Font("Arial", Font.BOLD, 24));
stockLabel.setFont(new Font("Arial", Font.BOLD, 24));
revenueLabel.setForeground(green);
activeLabel.setForeground(blue);
stockLabel.setForeground(red);
kpiPanel.add(createCard("TOTAL REVENUE", revenueLabel));
kpiPanel.add(createCard("ACTIVE ORDERS", activeLabel));
kpiPanel.add(createCard("LOW STOCK WARNINGS", stockLabel));
add(kpiPanel, BorderLayout.NORTH);
JTabbedPane tabs = new JTabbedPane();
createProductTab(tabs);
createOrderTab(tabs);
createCustomerTab(tabs);
add(tabs, BorderLayout.CENTER);
calculateKPI();
}

private JPanel createCard(String title, JLabel value) {
    JPanel card = new JPanel(new BorderLayout());
    card.setBackground(panel);
    JLabel titleLabel = new JLabel(title, SwingConstants.CENTER);
    titleLabel.setForeground(Color.LIGHT_GRAY);
    titleLabel.setFont(new Font("Arial", Font.BOLD, 13));
    card.add(titleLabel, BorderLayout.NORTH);
    card.add(value, BorderLayout.CENTER);
    return card;
}

private void createProductTab(JTabbedPane tabs) {
    JPanel mainPanel = new JPanel(new BorderLayout());
    mainPanel.setBackground(dark);
    String[] columns = {"Product ID", "Product Name", "Price ($)", "Stock Qty"};
    productModel = new DefaultTableModel(columns, 0);
    productTable = new JTable(productModel);
    styleTable(productTable);
    productModel.addRow(new Object[]{"P101", "High-End Laptop", 1200.00, 25});
    productModel.addRow(new Object[]{"P102", "Wireless Mouse", 35.50, 4});
    mainPanel.add(new JScrollPane(productTable), BorderLayout.CENTER);
    JPanel input = new JPanel(new GridLayout(2, 5, 8, 8));
    input.setBackground(panel);
    JTextField id = new JTextField();
    JTextField name = new JTextField();
    JTextField price = new JTextField();

```

```

        JTextField stock = new JTextField();
        JButton add = new JButton("Add Product");
        input.add(new JLabel("Product ID"));
        input.add(new JLabel("Name"));
        input.add(new JLabel("Price"));
        input.add(new JLabel("Stock"));
        input.add(new JLabel(""));
        input.add(id);
        input.add(name);
        input.add(price);
        input.add(stock);
        input.add(add);
        mainPanel.add(input, BorderLayout.SOUTH);
        add.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    if (id.getText().isEmpty() || name.getText().isEmpty() || price.getText().isEmpty()
|| stock.getText().isEmpty()) {
                        JOptionPane.showMessageDialog(EcommerceDashboard.this, "All fields are
required.");
                    }
                    return;
                }
                double p = Double.parseDouble(price.getText());
                int s = Integer.parseInt(stock.getText());
                productModel.addRow(new Object[]{id.getText(), name.getText(), p, s});
                id.setText("");
                name.setText("");
                price.setText("");
                stock.setText("");
                calculateKPI();
            } catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(EcommerceDashboard.this, "Enter valid
price and stock.");
            }
        });
        tabs.addTab("Products", mainPanel);
    }

    private void createOrderTab(JTabbedPane tabs) {
        JPanel mainPanel = new JPanel(new BorderLayout());
        mainPanel.setBackground(dark);
        String[] columns = {"Order ID", "Customer", "Product", "Amount ($)", "Status"};
        orderModel = new DefaultTableModel(columns, 0);
        orderTable = new JTable(orderModel);
        styleTable(orderTable);
        orderModel.addRow(new Object[]{"O501", "Alice Smith", "High-End Laptop", 1200.00,
"Processing"});
    }

```

```

        orderModel.addRow(new Object[]{"O502", "Bob Jones", "Wireless Mouse", 35.50,
"Shipped"});
        mainPanel.add(new JScrollPane(orderTable), BorderLayout.CENTER);
        JPanel input = new JPanel(new GridLayout(2, 6, 8, 8));
        input.setBackground(panel);
        JTextField old = new JTextField();
        JTextField custName = new JTextField();
        JTextField prodName = new JTextField();
        JTextField amount = new JTextField();
        JComboBox<String> statusCombo = new JComboBox<>(new String[]{"Processing",
"Shipped", "Delivered", "Cancelled"});
        JButton addOrder = new JButton("Add Order");
        input.add(new JLabel("Order ID"));
        input.add(new JLabel("Customer"));
        input.add(new JLabel("Product"));
        input.add(new JLabel("Amount ($)"));
        input.add(new JLabel("Status"));
        input.add(new JLabel(""));
        input.add(old);
        input.add(custName);
        input.add(prodName);
        input.add(amount);
        input.add(statusCombo);
        input.add(addOrder);
        mainPanel.add(input, BorderLayout.SOUTH);
        addOrder.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    if (old.getText().isEmpty() || custName.getText().isEmpty() ||
prodName.getText().isEmpty() || amount.getText().isEmpty()) {
                        JOptionPane.showMessageDialog(EcommerceDashboard.this, "All fields are
required.");
                        return;
                    }
                    double amnt = Double.parseDouble(amount.getText());
                    orderModel.addRow(new Object[]{old.getText(), custName.getText(),
prodName.getText(), amnt, statusCombo.getSelectedItem()});
                    old.setText("");
                    custName.setText("");
                    prodName.setText("");
                    amount.setText("");
                    statusCombo.setSelectedIndex(0);
                    calculateKPI();
                } catch (NumberFormatException ex) {
                    JOptionPane.showMessageDialog(EcommerceDashboard.this, "Enter a valid
order amount.");
                }
            }
        })
    }
}

```



```

});
tabs.addTab("Orders", mainPanel);
}
private void createCustomerTab(JTabbedPane tabs) {
    JPanel mainPanel = new JPanel(new BorderLayout());
    mainPanel.setBackground(dark);
    String[] columns = {"Customer ID", "Full Name", "Email", "Total Orders"};
    customerModel = new DefaultTableModel(columns, 0);
    customerTable = new JTable(customerModel);
    styleTable(customerTable);
    customerModel.addRow(new Object[]{"C301", "Alice Smith", "alice@email.com", 1});
    customerModel.addRow(new Object[]{"C302", "Bob Jones", "bob@email.com", 1});
    mainPanel.add(new JScrollPane(customerTable), BorderLayout.CENTER);
    JPanel input = new JPanel(new GridLayout(2, 5, 8, 8));
    input.setBackground(panel);
    JTextField cld = new JTextField();
    JTextField fullName = new JTextField();
    JTextField email = new JTextField();
    JTextField orderCount = new JTextField();
    JButton addCustomer = new JButton("Add Customer");
    input.add(new JLabel("Customer ID"));
    input.add(new JLabel("Full Name"));
    input.add(new JLabel("Email"));
    input.add(new JLabel("Total Orders"));
    input.add(new JLabel(""));
    input.add(cld);
    input.add(fullName);
    input.add(email);
    input.add(orderCount);
    input.add(addCustomer);
    mainPanel.add(input, BorderLayout.SOUTH);
    addCustomer.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
            try {
                if (cld.getText().isEmpty() || fullName.getText().isEmpty() ||
                    email.getText().isEmpty() || orderCount.getText().isEmpty()) {
                    JOptionPane.showMessageDialog(EcommerceDashboard.this, "All fields are
required.");

```

OUTPUT:

Applet Viewer: EcommerceDashboard.class

Applet

Dashboard System

TOTAL REVENUE
\$1235.50

ACTIVE ORDERS
2 Active

LOW STOCK WARNINGS
1 Items Low

Products

Orders

Customers

Product ID	Product Name	Price (\$)	Stock Qty
P101	High-End Laptop	1200.0	25
P102	Wireless Mouse	35.5	4

Add Product

Applet Viewer: EcommerceDashboard.class

Applet

Dashboard System

TOTAL REVENUE
\$1235.50

ACTIVE ORDERS
2 Active

LOW STOCK WARNINGS
1 Items Low

Products

Orders

Customers

Order ID	Customer	Product	Amount (\$)	Status
O101	John	High-End Laptop	1200.0	Active
O102	Emily	Wireless Mouse	35.5	Active

Add Order

Complete Order

Delete Order

Applet Viewer: EcommerceDashboard.class
Applet

Dashboard System

TOTAL REVENUE
\$1235.50

ACTIVE ORDERS
2 Active

LOW STOCK WARNINGS
1 Items Low

Products	Orders	Customers	
Customer ID	Customer Name	Email	City
C101	John	john@gmail.com	Chennai
C102	Emily	emily@gmail.com	Mumbai

Add Customer

Delete Customer